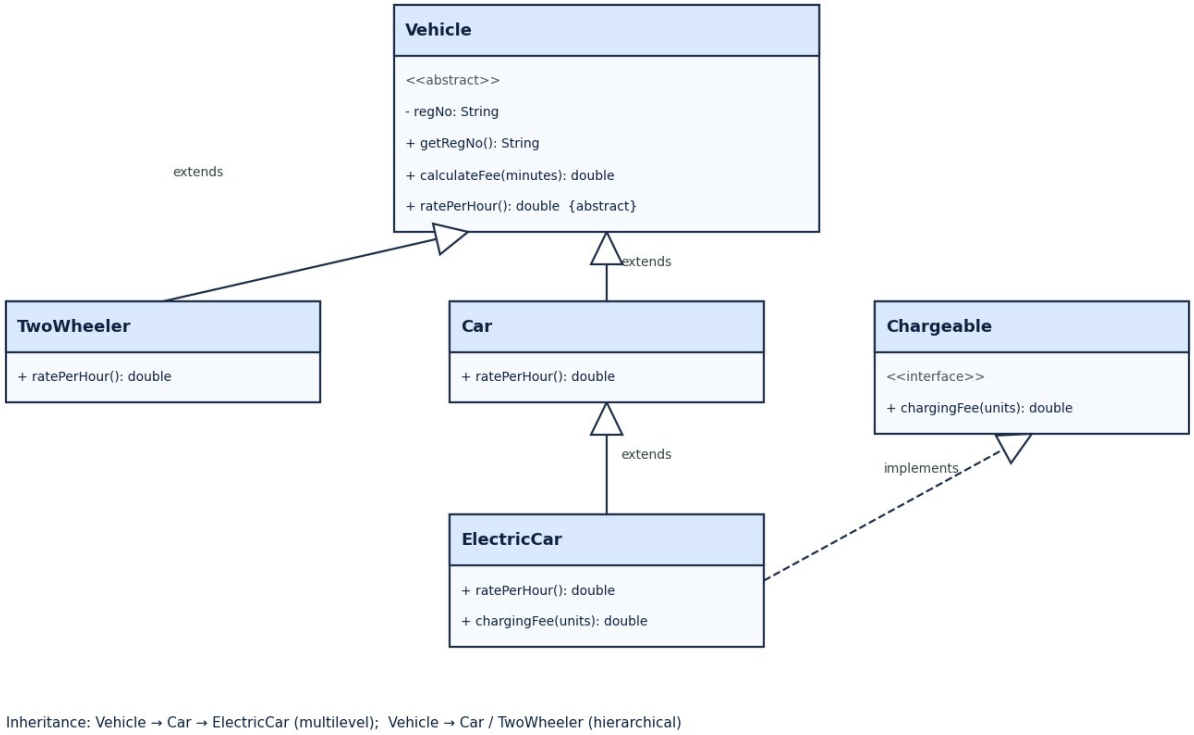
ASSIGNMENT 1 – TASK 6: Design the Inheritance Hierarchy

Bloom's Level: K4 – Analyse | Marks: 4

Use Vehicle as an abstract superclass. TwoWheeler and Car are direct subclasses; ElectricCar extends Car and implements Chargeable. This shows hierarchical inheritance (Vehicle has multiple direct subclasses) and multilevel inheritance (Vehicle → Car → ElectricCar). Java classes do not support multiple inheritance of classes, but a class may implement multiple interfaces.



super(...) calls the parent constructor. super.getRegNo() or super.someMethod() can call an accessible superclass method. A subclass inherits accessible fields/methods but not constructors. Keep fields private and use getters instead of accessing parent fields directly.

Constructor trace for new ElectricCar("TN03C3"): Java enters the ElectricCar constructor, which calls super(...); Car's constructor then calls Vehicle's constructor; the Vehicle constructor body runs first, then the Car constructor body, then the ElectricCar constructor body. Therefore the execution order of constructor bodies is Vehicle → Car → ElectricCar.

Composition alternative: Vehicle could contain a VehicleType or PricingStrategy object. Composition makes changing rates easier without creating a subclass for each pricing variation. Inheritance models an is-a relationship; composition models has-a. In either design, computing a simple fee is O(1) per vehicle and each object uses O(1) basic space.